

# Merged Exam Questions

1. QUESTION ONE (30 (30 MARKS).

2. a) Distinguish between knowledge and information (4 marks)

3. c) Explain three factors that affect the selection of a database management system.(6 marks)

4. c) State four types of keys that can be used in a relation. (4 marks)

5. e) Briefly describe three reasons for normalizing a database (6 marks)

6. f) Draw an E-R Diagram for Ruiru Hospital that captures the following entities: Patients. Doctors, Assistants and Wards. (5 marks)

7. g) State three database management systems. (3 marks)

8. QUESTION TWO ( (20 MARKS).

9. Question 2 (20 marks) a. Jack wants to design tables to be used in a given database. Using well labelled diagrams explain three types of entity relationships that he can use in the design. (10 marks) b. Models are used to represent the architecture of various systems. Discuss three types of database models that can be implemented in database design. (10 marks)

10. b) Models are used to represent the architecture of various systems. Discuss four types of database models that can be implemented in database design. (8 marks)

11. a) Explain five advantages of database management system over file management system. (5 marks)

12. a) Show, giving appropriate SQL commands, step by \_step how you would go about: i. Creating a database called Machakos with a table named Employee (attributes: Code, No, Name and Rate) in SQL. (4 marks) ii. Adding two records in each table. (4 marks) iii. Displaying names and rates employees (2 marks)

13. b) James wants to develop a database for a given set up. Discuss all the steps in the database development cycle that he is supposed to follow. (10 marks)

14. i) Discuss the process of designing a typical database. (6 marks) ii) Explain three properties of a good database design. (6 marks)

15. Question 4 (20 marks) a. Describe the five components of the DBMS environment and discuss how they relate to each other. (10 marks) b. Distinguish between the logical and physical database schema (4 marks) d. Erick, a database administrator is supposed to protect the database from various threats. Advise him on three methods he can use to perform this task. (6 marks)

16. b) Distinguish between the logical and physical database schema. (4 marks)

17. c) Erick, a database administrator is supposed to protect the database from various threats. Advise him on three methods he can use to perform this task. (6 marks)
18. b) Discuss three types of relationships that can be implemented in databases using well labelled diagrams. (6 marks)
19. QUESTION FIVE (20 MARKS) (a) Normalization is a technique that is used to organize tables so as to reduce redundancy. Briefly discuss the process of normalization from first up to the third normal forms (10 marks) (c) A systems administrator wants to organize files in the machines that exist in their work place. Explain five file organization methods that he may use. (10 marks)
20. c) Give the difference between a DDL and DML with examples in each case (4 marks)
21. i) Data integrity ii) Data Security (4 marks)
22. e) Briefly describe three (6 marks)
23. f) Draw an E-R Diagram for MUC that captures the following entities: Students. Lecturers, Library and Computer Lab. (5 marks)
24. g) State three roles of a database administrator. (3 marks)
25. b) Show, giving appropriate SQL commands, step by step how you would go about the following tasks:
26. i) Creating a database called Macha with two tables named Project (attributes: Code, Title, Manager, Budget) and Employee (attributes: Code, No, Name and Rate) in MySQL. (4 marks) ii) Adding two records in each table. (4 marks) iii) Displaying names and rates of employees (2 marks)
27. a) Using real world examples, Explain the following terms as used in database systems:- i. Entity (3 marks) ii. Entity Type (3 marks) iii. Entity Set (3 marks)
28. b) Describe three levels of abstraction in the DBMS? (6 marks)
29. c) Highlight three types and use and application of various Relational Database Keys (6 marks)
30. d) Describe :- i. The best practices in database security that can help keep your databases safe from attackers. (4 marks) ii. The properties of second normal form (5 marks)
31. a) Define a CLAUSE in terms of SQL (2 marks)
32. e) Describe THREE Database Integrity rules and provide example for each (6 marks)
33. c) Explain two disadvantages of ER Model in DBMS (4 marks)
34. d) Discuss the characteristics of a relational table (8 marks)

35. c) Distinguish between a distributed database and a distributed database management system. (4 marks)
36. b) Using a relational illustration, explain functional dependency in the DBMS (3 marks)
37. c) Describe the Importance of Data Modeling in DBMS (4 marks)
38. a) Define a RDBMS (2 marks)
39. b) Explain three characteristics of Network Database Management System (DDBMS) (8 marks)
40. d) Analyse four states of transaction management in DBMS (8 marks)
41. a) List four roles of a database administrator (4 marks)
42. b) Given user Brian and user Ben are transacting and updating record in the same table, with the aid of a diagram, illustrate a lost update as a result of an inconsistent read. (12 marks)
43. c) Discuss two Database authentication methods in a Local Area Connection. (4 marks)
44. d) While setting REMOTE\_LOGIN\_PASSWORDFILE, several aspects must be taken into account. Discuss three which can be set on this parameter and their implication. (6 marks)
45. e) In a busy network operation center, there have been several cases of security breach. You have been hired as a consultant to help settle the matter. Discuss three approaches you would use to seal the loopholes. (6 marks)
46. f) You have been hired as a database administrator in a data center. The center is being set up for the first time. Highlight four factors you would consider while creating a database. (4 marks)
47. a) Enumerate four key security features which every database management software should have. (4 marks)
48. b) While creating a database, and administrator can use various methods. Discuss three such methods (6 marks)
49. b) Explain the role of the following in integrity control (10 marks) i. Views. ii. Integrity Controls: iii. One type Domain. iv. Assertions. v. Triggers.
50. a) List three causes of flawed user authentication. (3 marks)
51. b) Propose two methods to solve user authentication flaws in database management systems. (4 marks)
52. c) List four database system actions which can be assigned to a subject to execute on an object. (4 marks)

53. d) List four Mechanisms to facilitate restoration of a database quickly and accurately after loss or damage (2 marks)
54. e) While creating a database, DBCA is very useful in some circumstances. Discuss three advantages of using DBCA in database creation. (6 marks)
55. a) Explain five back up facilities in a database management system. (10 marks)
56. b) Discuss the journalizing aspect of a database system where it creates an audit trail. (10 marks) (4 marks)
57. c) Database performance is determined by several parameters. Determine four aspects of database system that a database administrator needs to monitor. (4 marks)
58. d) Define the following terms as they relate to database administration (6 marks) i. Tablespace ii. Instance iii. Data dictionary iv. Redo log v. Schema vi. Timestamp
59. a) Databases are bound to fail due to various factors. Discuss four options that a database administrator can choose to respond to the failure, Indicating the circumstances most suitable for each. (8 marks)
60. b) Discuss the following terms concurrency control and Serialization as applied database management (4 marks)
61. c) Highlight four levels of locks. (4 marks)
62. d) With the aid of a diagram, show how roll forward database recovery works. (4 marks)
63. a) Distinguish between the following terms and concepts. (4 marks) (i) Data Physical Format and Data Logical Format as applies to data dependence. (ii) Data Inconsistency and Data Anomalies as applies to data redundancy. b) "The Database Administrator (DBA) must be prepared to recover data to a usable point, no matter what the cause, and to do so as quickly as possible.". Explain the three types of data recovery referred to in this statement. (6 marks)
64. c) When asked by their lecturer why they think studying relational algebra and relational calculus is important in Advance Database Systems, the following students responded as follows. (4 marks) Janet Relational Algebra provides the relational model with a flexible way to query a database. Brian Relational Calculus is the foundation of Query-By-Example. Justify these statements.
65. d) Outline three common characteristics that data models must have in order to be widely accepted. (3 marks) e) "When implementing a physical database from a logical data model, there is need to consider how the database will perform when applications make requests to access and modify data.". Discuss this statement with emphasis on techniques that are employed to allow data to be accessed in the database more rapidly. (4 marks)
66. f) Explain two ways in which views can be used to implement data security. (2 marks)
67. a) Outline two advantages and two disadvantages of object-oriented database model. (4

marks)

68. c) Discuss five reasons why you would advise a company to go for a database management system other any other data management system. (5 marks)

69. i) Explain any two important functions it performs that guarantee the integrity and consistency of data in the database. ii) One technical and one non-technical DBMS strategy an organization can adopt in establishing a usable database environment. iii) Two considerations needed in understanding the DBMS requirements and preparation of the environment for the new DBMS in the organization. iv) One benefit and one risk in upgrading the DBMS in organizations database environment.

70. c) Consider the relation R(A,B,C,D,E,F,G,H,I,J,K) which is in First Normal Form (1NF). Suppose its dependencies are A,B!'C B,D!'E,F A,D!'G,H A!'J H!'K.

71. i) Identify the key of R. (2 marks) ii) Decompose R into 2NF relations. (3 marks) iii) Then decompose R into 3NF relations. (3 marks)

72. a) The following are relations from a database. Use them to answer the questions that follow. (10 marks) Sailors (sid, sname, rating, age) Boats (bid, bname, color) Reserves (sid, bid, day) Represent the query below in Relational Algebra, Tuple Relational Calculus, Domain Relational Calculus and Structured Query Language (SQL). Find the names of sailors who reserved boat #103.

73. c) Write the following SQL statements to perform the following

74. i) Updating one or more records (2 marks) ii) Changing attributes characteristics (2 marks) iii) Deleting all rows in from a table (2 marks) iv) Primary key designation (2 marks)

75. b) Explain the significance of the two basic types of Data Control Language statements as used in database security. (4 marks)

76. c) Using an SQL statement, explain two types of privileges that can be granted and revoked from database users. (4 marks)

77. d) Write SQL statements that demonstrate the creation of a trigger and a stored procedure in databases. (4 marks)

78. f) In a transaction, explain three aspects that a DBMS should guarantee with respect to concurrent execution of several transactions and database consistency (6 marks)

79. b) Consider database objects A and B and assume that there are two transactions T1 and T2. Transaction T1 reads objects P and Q and then writes object P. Transactions T2 reads object P and Q and then writes objects P and Q. State an example schedule with actions of transactions T1 and T2 on objects P and Q that results in a write-read conflict. (4 marks)

80. c) Discuss the role of a database administrator enforcing security in database design. (4 marks)

81. d) Explain the difference between the two methods used for accessing relational data from a Java program. (4 marks)
82. a) Explain how the following database concepts are implemented i. Data availability (4 marks) ii. Data validation (4 marks) iii. Data security (2 marks)
83. b) A tennis club uses the following table to record details of players and their coaches.  

| PlayerID | Name    | Ranking | CoachID | CoachName |
|----------|---------|---------|---------|-----------|
| P001     | Kariuki | 12      | C003    | Ben       |
| P002     | Okoth   | 3       | C013    | Mark      |
| P003     | Tanui   | 17      | C003    | Ben       |
| P004     | Mosioma | 9       | C006    | Titus     |

i. Explain why the above table is not in 3rd normal form. (4 marks) ii. Transform the table into 3rd normal form. (6 marks)
84. c) The table below records orders for items. Each order is placed on a given date, and may include a variety of items in different quantities. Orders  

| OrderNo | ItemNo | Description | Date    | Quantity |
|---------|--------|-------------|---------|----------|
| 1       | 12     | Screw       | 6th jan | 100      |
| 1       | 15     | bolt        | 6th jan | 50       |
| 2       | 7      | Flange      | Feb 2nd | 10       |
| 2       | 15     | Bolt        | Feb 2nd | 2        |
| 40      | 2      | 12 screw    | Feb 2nd | 80       |

i. Give an example of an insertion anomaly and an example of modification (update) anomaly that might occur in the above table. (4 marks) ii. Transform the table into 2nd Normal Form (6 marks)
85. a) Explain three stages of data base design (6 marks)
86. a) Describe three integrity constraints that could be applied to a database management system during its design stage. (6 marks)
87. b) Explain two update anomalies associated with poor database design. (2 marks)
88. c) Sharif is a newly recruited database administrator. Explain four roles that he would be expected to perform in his everyday operations (6 marks)
89. d) Describe five advantages of using a database system over the use of spreadsheets. (6 marks)
90. a) Highlight three characteristics of using a primary key during database design (6 marks)
91. a) Describe, using an example, one type of problem that can occur in a multi-user environment when concurrent access to the database is allowed. (10 marks)
92. b) By using illustrations, state the difference between hierarchical and network database models (10 marks)
93. a) Distinguish between a field and record as used in database systems (4 marks)
94. c) Explain five reasons for having a database model. (10 marks)
95. i) Relational Model (3 marks) ii) Object oriented model (3 marks)
96. d) Explain the following phases of database development
97. i) Evaluation and Selection ii) Requirements Analysis iii) Implementation (6 marks)

98. f) Draw an entity relationship diagram for Machakos level five hospital that captures any four entities. (4 marks)

99. a) The table below represents details of an entity to be modelled in a database. Use it to answer the questions that follow. Order Date Customer Items 22/8/2021 Mark Okode Computer, Printer, Pen Normalize this table into:

100. i) First Normal Form (4 marks) ii) Second Normal Form (3 marks) iii) Third Normal Form (3 marks)

101. b) Explain three differences between Oracle DBMS and SQL database. (6 marks)

102. c) Distinguish between data definition language and data manipulation language as used in SQL. (4 marks)

103. a) Database design is an important stage in development:

104. b) James has been asked to create a relationship between two tables in MS-Access. Explain the process he is likely to follow to accomplish this task. (4 marks)

105. c) State four categories of database users (4 marks)

106. b) Schema is a way of logically grouping database objects. Explain three types of database schema that can be used for this purpose. (6 marks)

107. a) Explain the function of each of the clauses in the SELECT statement in SQL

108. i) From ii) Where iii) Group By (6 marks)

109. a) Show, giving appropriate SQL commands, step by \_step how you would go about the following tasks: i. Create a database called Macha Uni. (2 marks) ii. Select the database above and create a table called Staff with five fields. Include the appropriate data types, primary key and other attributes. (5 marks) iii. Add two records in the table. (4 marks) iv. Select all Female Lecturers in ICT department (4 marks)

110. a) List four competencies which a database administrator should have to do his work effectively. (4 marks)

111. c) As an organization grows, there is need to establish a data warehouse. State four key roles of a data ware house administrator. (4 marks)

112. a) Discuss four database failure responses under the following circumstances. (8 marks) i. Aborted transactions ii. Incorrect data iii. System failure (database intact) iv. Database destruction

113. d) Differentiate shared lock and an exclusive lock. (4 marks)

114. c) Discuss four components of a database environment (4 marks)

115. d) Using clear examples, illustrate the application of three sql languages (6 marks)
116. e) Illustrate two approaches of updating data in data to a table in Ms. Access (4 marks)
117. b) Study the following table and answer the questions that follow.
118. i) Normalize the above table to the 3NF (6 marks) ii) State the three main differences between 1NF, 2NF and 3NF tables. (3 marks)
119. c) Out of the 3NF table(s), generate a SQL script that displays student number, name, campus, department, and class using an equijoin. (7 marks)
120. a) Justify the importance of adding indexes in a table (2 marks)
121. b) List the four phases involved in query processing (4 marks)
122. c) Study the following table and answer the following questions: P\_ID F\_Name S\_Name P\_Category P\_DOB Tel \_Number M\_Status HomeAddress Write the SQL script to do the following:
123. i) Create the above table (4 marks) ii) Inserts these values in the table (1 , John, Peterson, Adult, 12/12/1970, 0111123, Single, 233-2250 Narok) while applying lock to the entire table (3 marks) iii) Displays all patients below 18 years (3 marks) iv) Adds a constraint to avoid repetition of PatientID (4 marks)
124. a) List six MS access (3 marks)
125. b) Justify the importance of a database administrator in an ERP system (4 marks)
126. c) Highlight four actions involved in a checkpoint in a database system. (4 marks)
127. a) Define concurrency in a database transaction (2 marks)
128. b) Simulate the following occurrences in a database system.
129. i) Lost Update (4 marks) ii) Uncommitted dependency (4 marks) iii) Inconsistent analysis (4 marks) c). Discuss three roles of concurrency control (6 marks)
130. a) List five database management system software used in industry today (5 marks)
131. e) Illustrate the step by step process of generating a report from Ms access table. (4 marks)
132. c) Discuss the three levels of schema architecture. (6 marks)
133. d) Differentiate the following (4 marks) i. Logical and physical data independence ii. Single user and multi user systems.
134. c) Study the following table and answer the following questions: Index PatientID PatientName PatientCategory DOB MaritalStatus DisabiltyType HomeCounty Write the SQL



script to do the following:

135. a) Define a database model (2 marks)

136. b) Study the following table and answer the questions that follow. Std\_no Name, Tel\_No Campus, Town Department HoD Office\_Tel  
401 Jane, 0721545 Main, Kenya CIT Mr. X 53337  
402 Kamau, 077120 Main, Kenya ENG Mr. Y 53341  
403 Kioko, 0721125 Main, Nairobi CIT Mr. X 53337  
404 Lenku, 072564 Town, Kericho ENG Mr. Y 53337  
i). Normalize the above table to the 3NF showing all the proceeding forms (15 marks)  
ii). State the three main differences between 1NF, 2NF and 3NF tables (3 marks)

137. c) Give the difference between delete anomaly and update anomaly. (4 marks)

138. d) Explain the following database concepts i. OLTP ii. OLAP (4 marks)

139. e) Briefly describe three possible attacks to a database system (6 marks)

140. f) Draw an Entity Relationship Diagram for MU that captures the following entities: Students. Lecturers, Library and Computer Lab. Include 2 attributes for each entity. (4 marks)

141. b) Explain four file organization methods that can be used in databases. (8 marks)

142. a) Distinguish between the following: i. Primary key and foreign key ii. Inner join and right join (8 marks)

143. c) Discuss three constraints that can be used in database tables (6 marks)

144. a) Explain the following terms as used in database systems:- i. Atomicity (2 marks) ii. Aggregation (2 marks) iii. ERD (2 marks)

145. b) Define 'Data Independence (2 marks)

146. e) Describe three objectives of the selection of a data type for databases. (6 marks)

147. f) Write an SQL SELECT statement to display all the columns of the STUDENT table but only those rows where the Grade column is greater than or equal to 90. (3 marks)

148. g) Explain what needs to happen to convert a relation to third normal form (6 marks)

149. a) Write an SQL SELECT statement to count the number of rows in STUDENT table and display the result with the label NumStudents. (3 marks)

150. b) Discuss the alternative terminology that is used in the relational model. (4 marks)

151. c) Highlight the steps that should be followed when preparing to create a database table (8 marks)

152. d) In database management systems; i. Define domain constraints (1 mark) ii. Describe TWO domain constraints (4 marks)

153. a) Explain why normalized tables require more complex SQL when SQL statements are used in application programs (4 marks)

154. b) Explain four Problems with Traditional File System in Data Management Processing (4 marks) Study the following database table that stores student's records and answer the questions that follow:-

| REG_NO | F_NAME | L_NAME   | COURSE | GENDER | D_O_B      |
|--------|--------|----------|--------|--------|------------|
| 1/2016 | John   | Mutuku   | DCS    | Male   | 13/6/1990  |
| 2/2016 | Steve  | kipkorir | DCS    | Male   | 13/3/1985  |
| 3/2016 | Susan  | Mutua    | CIT    | Female | 19/11/1986 |
| 4/2016 | Steve  | Kin'gori | DBIT   | Male   | 1/3/1978   |

155. i) Display all students where the results will only contain the columns RGE\_NO, L\_NAME and F\_NAME. (2 marks) ii) To produce the L+NAME and F\_NAME combined as Student name (3 marks) iii) To find the details of All male students in the course DCS. (2 marks) iv) To produce the total number of students per course (2 marks)

156. v) Get a list of all the male students born between the year 1980 and 1989. (3 marks)

157. b) Describe the difference between homogeneous and heterogeneous distributed database (4 marks)

158. c) Explain TWO Applications of Distributed Databases (2 marks)

159. d) You have two tables, EMPLOYEE and COMPUTER that are in a one-to-one relationship. The foreign key is EmpNumber in COMPUTER which references EmpNumber as the primary key of EMPLOYEE. Explain what must be done to convert the one-to-one EMPLOYEE-COMPUTER relationship to a one-to-many relationship where one employee can have more than one computer. (6 marks)

160. b) The Database Life Cycle (DBLC) contains six phases, Analyze what happens in phase one. (6 marks)

161. c) Discuss FOUR disadvantages of a standard language such as SQL (8 marks)